# Platform Architecture & Execution Lifecycles

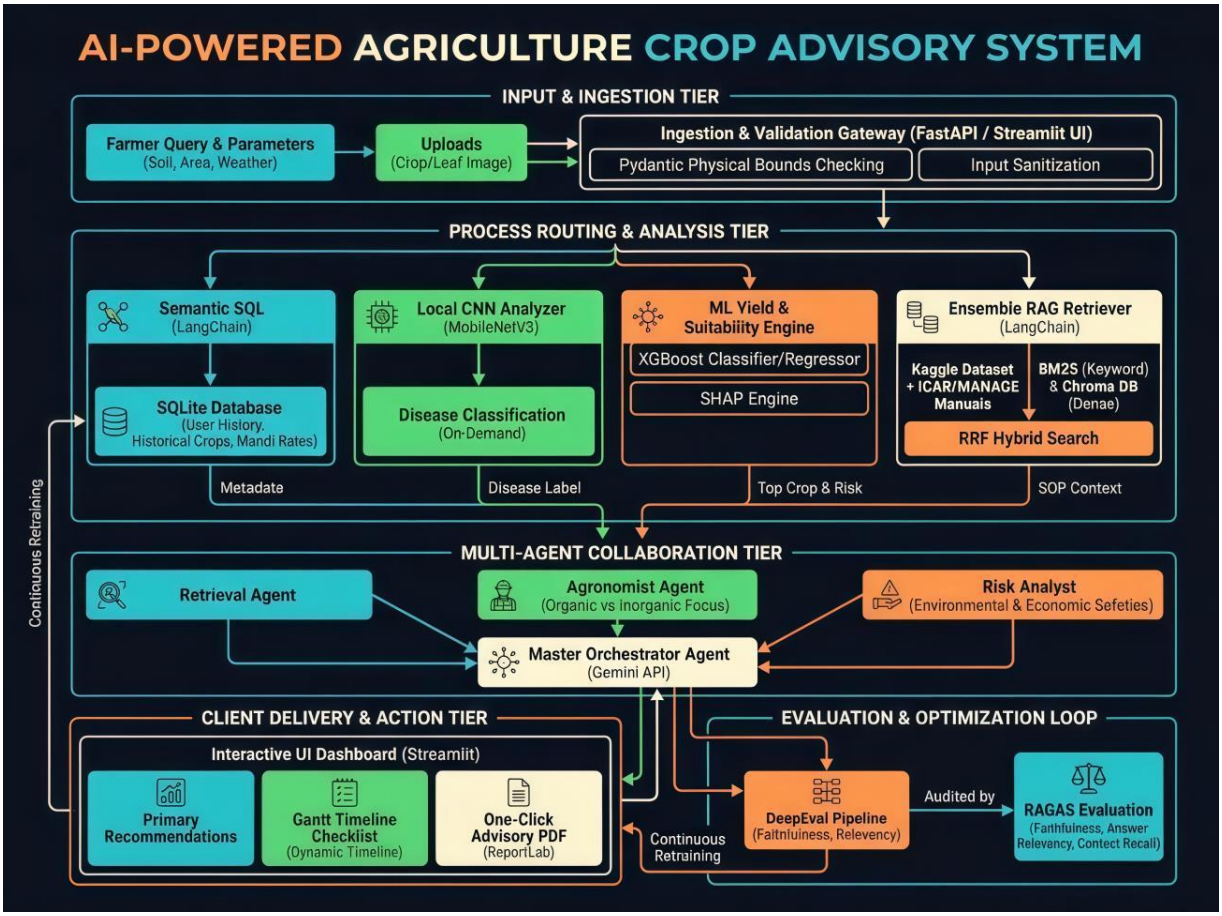
Visual Architectural Diagrams with Contextual Technical Descriptions

SYSTEM CONTEXT & STATIC ARCHITECTURE

## End-to-End System Architecture Topology

**Overview:** This diagram illustrates the decoupled five-tier system architecture connecting farmer input channels to local analytical engines, multi-agent LLM orchestrators, and automated quality benchmarks:

- **Input & Ingestion Tier:** Captures field parameters (N, P, K, pH, weather) and leaf photographs via Streamlit and FastAPI with strict Pydantic physical bounds validation.
- **Process Routing & Analysis Tier:** Concurrently runs the ML Suitability Engine (ExtraTrees + SHAP), Hybrid RAG Retriever (ChromaDB + BM25 over ICAR manuals), Read-Only SQL Agent (SQLite), and CNN Leaf Doctor (MobileNetV3).
- **Multi-Agent Collaboration Tier:** Combines specialized Agronomist and Risk Analyst agents with a Master Gemini Orchestrator using Reciprocal Rank Fusion (RRF).
- **Client Delivery Tier:** Streams interactive UI recommendation cards, 5-stage cropping calendars, and publication-grade ReportLab PDF dossiers.
- **Evaluation & Optimization Loop:** Continuously benchmarks faithfulness, relevancy, and safety using DeepEval and RAGAS.

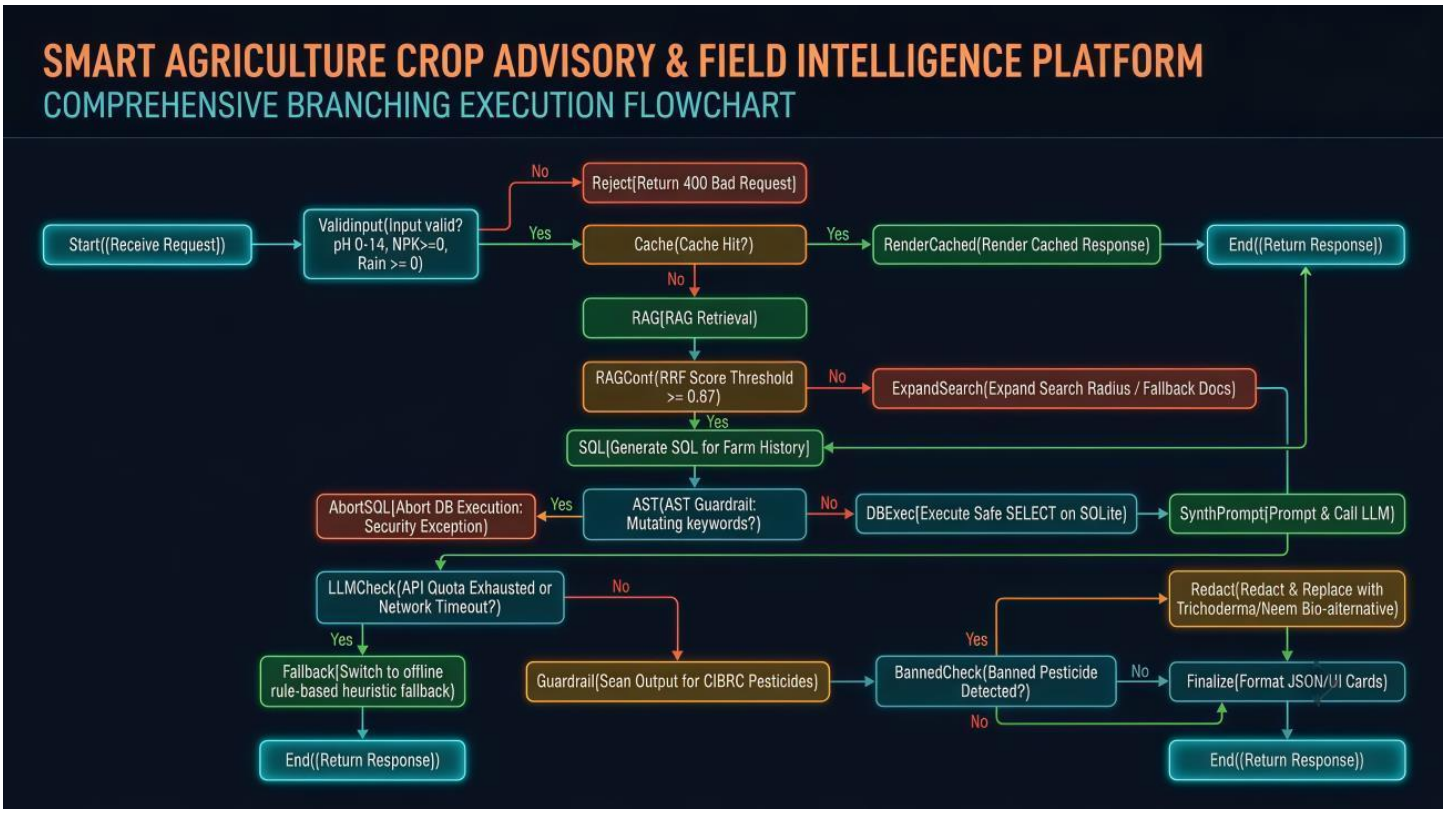


RUNTIME DECISION & GUARDRAIL LOGIC

## Comprehensive Branching Execution Flowchart

**Overview:** This execution flowchart maps the conditional routing, security checks, and fallback mechanisms executed for each advisory request:

- **Input Validation Gate:** Verifies physical boundaries (pH 0–14, positive rainfall/NPK); immediately returns HTTP 400 Bad Request on invalid data.
- **Cache & Fast-Path Inference:** Bypasses heavy processing on cache hit to deliver immediate sub-50ms responses for warm queries.
- **Hybrid RAG & AST Safety:** Scores retrieved research chunks using Reciprocal Rank Fusion (RRF threshold 0.67); inspects SQL queries via an AST validator to block any mutating statements (DROP/DELETE/UPDATE).
- **Quota Intercept & Offline Heuristic:** Detects LLM quota exhaustion or network timeouts and immediately activates an offline rule-based heuristic fallback.
- **CIBRC Pesticide Scanner:** Automatically scans generated text for statutory banned agrochemicals (Endosulfan, Paraquat), redacting violations and substituting organic bio-alternatives (Neem/Trichoderma).

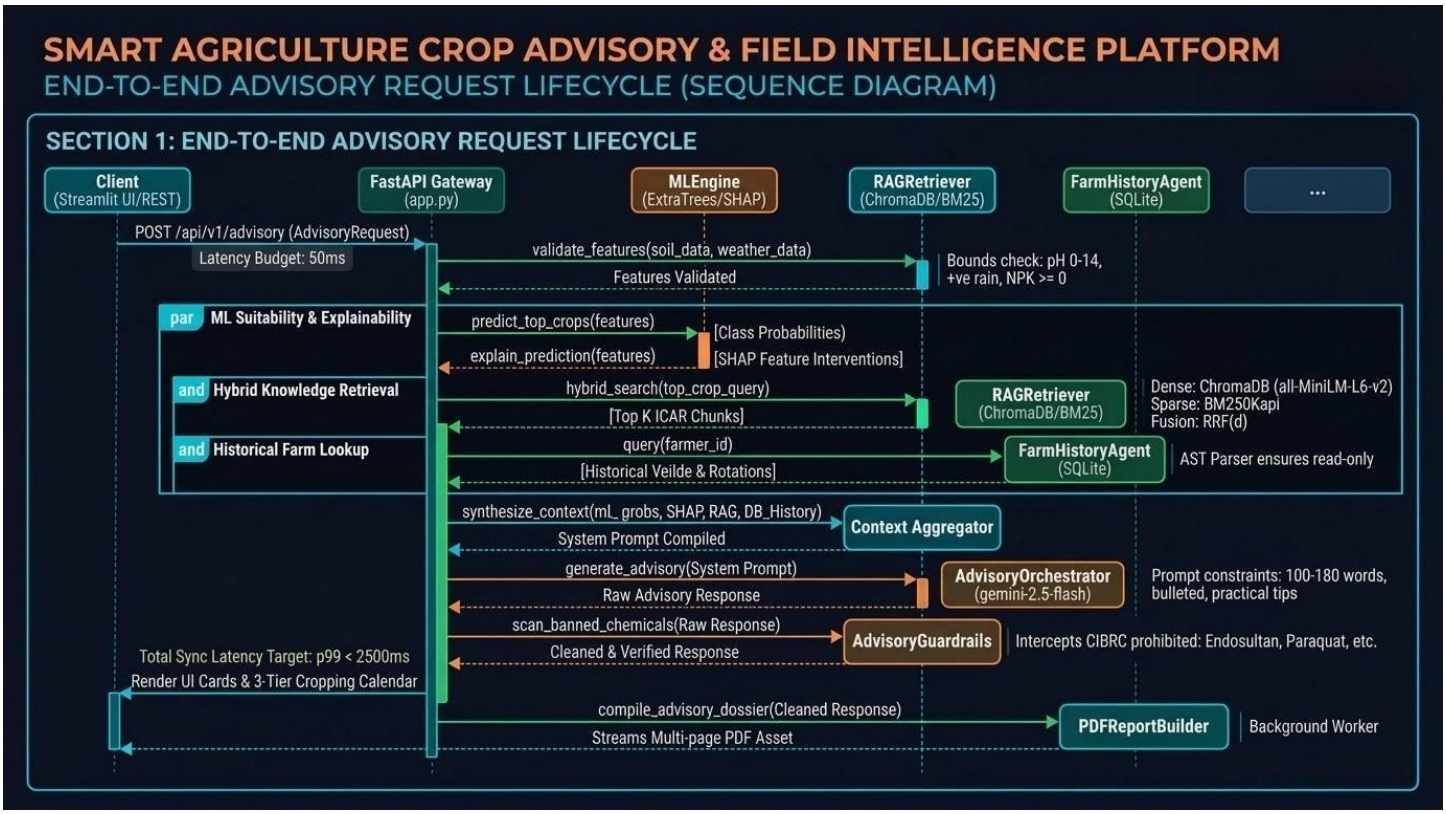


RUNTIME COMMUNICATION & LATENCY BUDGET

## End-to-End Advisory Request Lifecycle (Sequence Diagram)

**Overview:** This sequence diagram documents component interaction timelines and latency budgeting for a complete advisory lifecycle:

- **FastAPI Gateway Ingestion:** Receives POST /api/v1/advisory and runs initial soil and weather parameter checks.
- **Parallel Engine Dispatch:** Concurrently triggers MLEngine (ExtraTrees crop suitability + SHAP feature attributions in <50ms), RAGRetriever (dense ChromaDB + sparse BM25), and FarmHistoryAgent (SQLite historical yields).
- **Prompt Synthesis & Constraints:** AdvisoryOrchestrator packs aggregated context into a token-budgeted prompt enforcing farmer-brevity rules (100–180 words, bulleted action items).
- **LLM Generation & Safety Redaction:** Calls Gemini LLM, routes raw response through AdvisoryGuardrails to intercept prohibited chemicals, and delivers sanitized output.
- **Dossier Rendering:** Renders UI cards and dispatches background worker PDFReportBuilder to stream a multi-page PDF dossier with target p99 latency < 2,500ms.

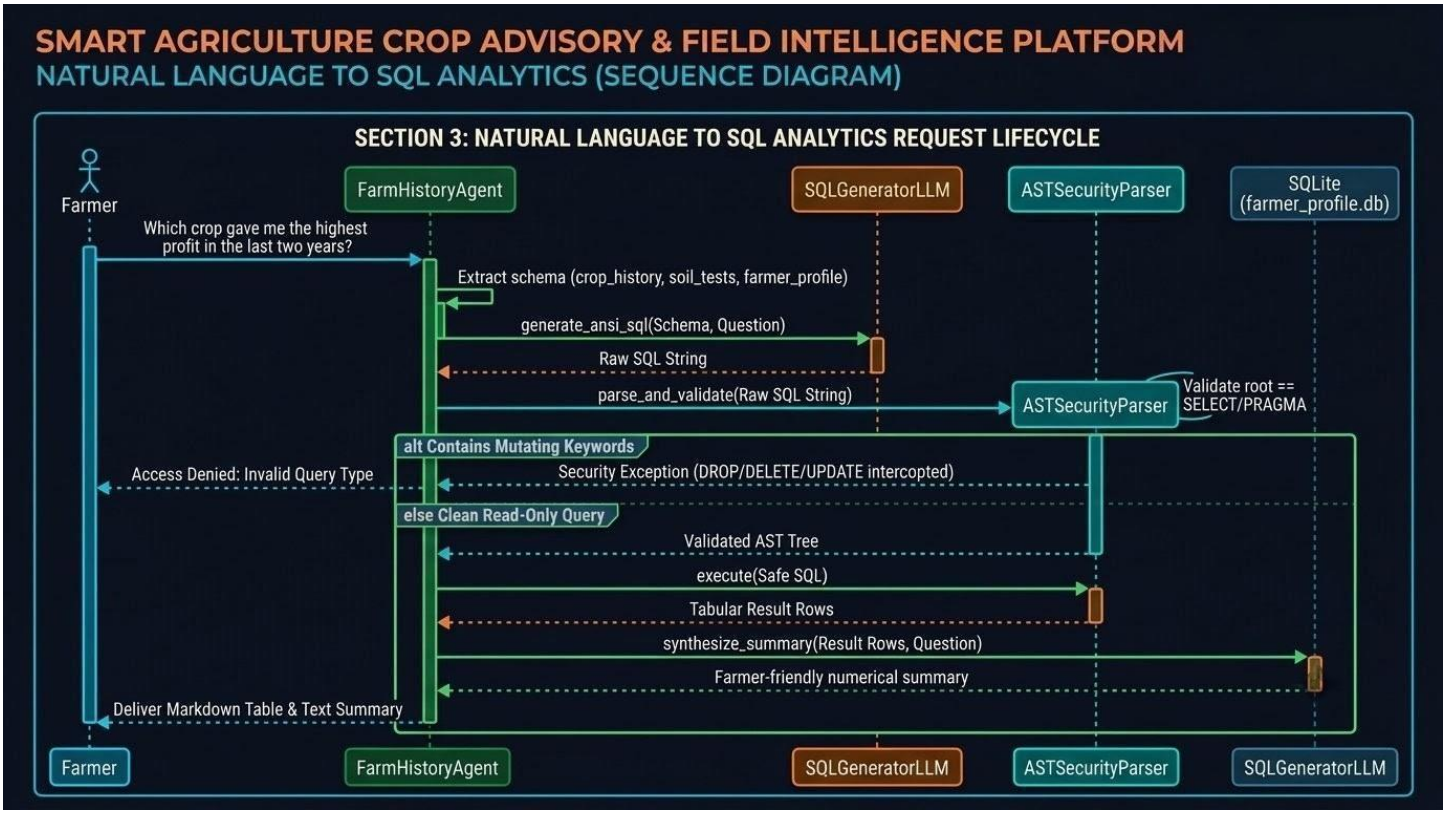


DATABASE QUERY & AST SECURITY PROTOCOL

## Natural Language to SQL Analytics Lifecycle (Sequence Diagram)

**Overview:** This sequence diagram outlines how natural language queries over historical farm records are validated, executed, and summarized:

- **Farmer Question Ingestion:** Farmer asks natural questions regarding past harvests (e.g. *'Which crop gave me the highest profit in the last two years?'*).
- **Schema Binding & SQL Generation:** FarmHistoryAgent extracts schema definitions (crop_history, soil_tests, farmer_profile) and prompts the LLM to generate ANSI SQL.
- **Read-Only AST Security Parser:** Inspects the Abstract Syntax Tree (AST); immediately blocks mutating statements (DROP, DELETE, UPDATE, ALTER) with an Access Denied security exception.
- **Safe Execution on SQLite:** Dispatches verified read-only SELECT queries against local database farmer_profile.db.
- **Response Synthesis:** Synthesizes raw database result rows into farmer-friendly numerical summaries and formatted Markdown tables.



FINITE STATE MACHINE ARCHITECTURE

## System State Transition & Decision Lifecycle (State Diagram)

**Overview:** This finite state machine diagram maps the runtime operational states and error recovery pathways across request execution:

- **Idle_AwaitingInput → ValidatingInput:** Transitions on request receipt. Invalid features trigger FeatureRejected for client correction.
- **ParallelProcessing:** Concurrently executes ModelInference (ML), HybridRetrieval (RAG), and SQLContextLookup (DB history).
- **PromptSynthesis → LLMGenerating:** Compiles aggregated telemetry into the system prompt. Network or quota failures trigger FallbackMode to serve local agronomic heuristics.
- **GuardrailVerification:** Scans generated output. CIBRC violations transition to BannedChemicalDetected for redaction and bio-alternative substitution.
- **FormattingResponse → Completed:** Formats final UI cards and cropping calendar milestones while initiating asynchronous PDFRendering in the background.

